README: The MultiWiSE Canada Dashboard

Purpose

This dashboard uses information on individual-level residential histories (6-character postal codes) and defined exposure windows to calculate the 12 Multiyear Wildfire Smoke Exposure (MultiWiSE) metrics, which characterize an individual's episodic exposure to wildfire smoke (WFS) $PM_{2.5}$. The dashboard can be used to generate $PM_{2.5}$ exposure profiles and the MultiWiSE metrics for multiple individuals residing in Canada for any period between 2010-2023. Details on the required inputs to the dashboard can be found below in 'Usage', and details on the outputs provided by the dashboard can be found below in 'Outputs.' More information on the datasets and methods used to generate the MultiWiSE metrics can be found in the manuscript '*Multiyear Wildfire Smoke Exposure (MultiWiSE) metrics: A data-driven approach to characterizing episodic $PM_{2.5}$ exposures for epidemiologic research.*' **Figure 1** outlines the step-by-step approach the dashboard uses to generate an individual's exposure profile and MultiWiSE metrics.

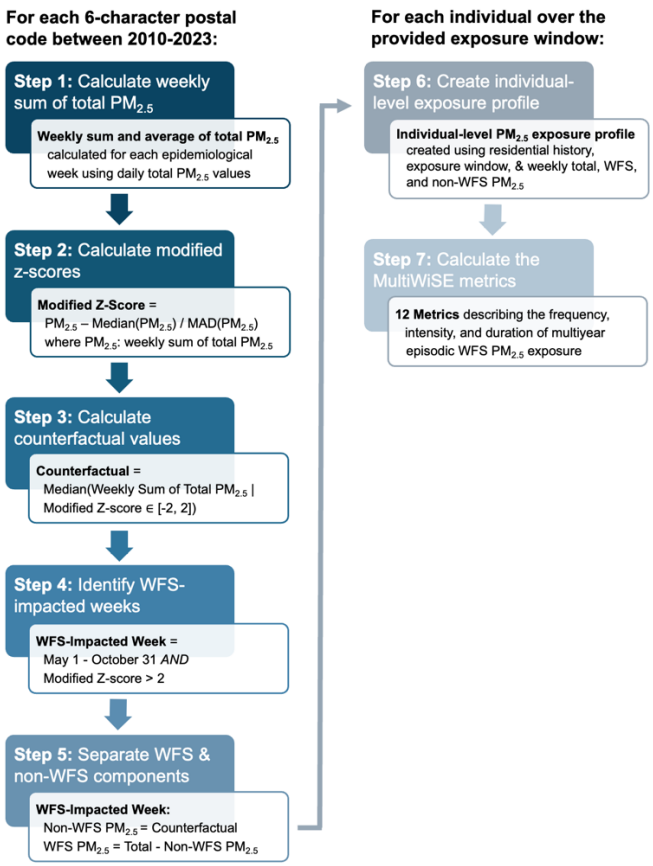


Figure 1. The step-by-step methodological approach used in the dashboard to calculate the 12 Multiyear Wildfire Smoke Exposure (MultiWiSE) metrics. Steps 1-5 generate the weekly $PM_{2.5}$ estimates that are used in Steps 6-7. Steps 6-7 are applied separately to each individual.

Usage

This dashboard uses pre-generated estimates of weekly total (i.e., all-source), wildfire smoke (WFS), and non-WFS $PM_{2.5}$ across Canada for 2010-2023 (publicly available via Dropbox - see 'Supplemental' below), in combination with user-provided residential histories and exposure windows, to generate individual-level $PM_{2.5}$ exposure profiles and the MultiWiSE metrics.

To use the dashboard, the following information must be provided for each individual:

- The exposure window (i.e., time period over which you want to calculate the individual's exposure, which must be at least 1 year in length)
- The postal code(s) the individual lived in for the duration of the exposure window, and the date(s) they moved into and out of each postal code

This information should be provided as two separate files (.csv, .xlsx, or .xls format), one for residential history and one for exposure window. **Table 1** below outlines the required fields and formatting of the .csv files, and **Table 2** and **Table 3** provide examples of the required files:

- **File 1:** Individual-Level Residential History
  For each individual, one row per postal code they lived in during the exposure window, with information on the dates they moved into and out of each postal code. Required columns include: `IndividualID, PostalCode, StartDate, EndDate`.
- **File 2:** Individual-Level Exposure Window
  One row per individual with the start and end date of the exposure window of interest. This exposure window can be different for each individual. Required columns include: `IndividualID, ExposureStart, and ExposureEnd`

Table 1. Descriptions of the Required Variables

Variable	Type	Description
File 1: Individual-Level Residential History		
IndividualID	Character string	Unique identifier for individual
PostalCode	Character string	6-digit alphanumeric Canadian postal code with no spaces
StartDate	Date (YYYY-MM-DD)	Date individual moved into the postal code
EndDate	Date (YYYY-MM-DD)	Date individual moved out of the postal code
File 2: Individual-Level Exposure Window		
IndividualID	Character string	Unique identifier for individual
ExposureStart	Date (YYYY-MM-DD)	Start date of the exposure window
ExposureEnd	Date (YYYY-MM-DD)	End date of the exposure window

Table 2. Example file for residential histories

IndividualID	PostalCode	StartDate	EndDate
Person1	Y1A0J4	2012-01-27	2016-04-31
Person1	T8V8K9	2016-05-01	2022-11-15
Person2	C0A0A1	2010-03-15	2013-06-18
Person2	C0A0A2	2013-06-19	2018-12-31
Person2	B3M3K2	2019-01-01	2023-10-15
Person3	V0E0A3	2011-07-01	2023-07-01

Table 3. Example file for exposure windows

IndividualID	ExposureStart	ExposureEnd
Person1	2014-07-01	2022-07-31
Person2	2012-11-15	2023-08-01
Person3	2011-07-01	2023-07-01

For the dashboard to work as expected, the files should meet the following conditions:

- Column names must be exact: `IndividualID, PostalCode, StartDate, EndDate` in the residential history file, and `IndividualID, ExposureStartDate, ExposureEndDate` in the exposure window file
- All dates must be written as YYYY-MM-DD (e.g. 2022-07-15)
- The provided postal codes must be valid and located in one of the 13 provinces or territories in Canada. The postal codes must be formatted to have 6 digits and no spaces (e.g., V5Z4R4)
- Each individual must appear in both the exposure window file and the residential history file
- The residential history for an individual should be contiguous (i.e., no gaps) and non-overlapping, and the start date for a given postal code must occur before the end date
- The exposure window for each individual must fall within the range of available data (January 1, 2010 – December 31, 2023), be at least 1 year in length, and the exposure window start date must occur before the end date
- The residential history for an individual must include residential information for the entirety of the exposure window
- Each individual must appear exactly once in the exposure window .csv file. Multiple exposure windows for the same IndividualID will not be processed
- There can be no blank cells
- Files provided as a .csv, .xls, or .xlsx

If any of the above conditions are not met, the dashboard will either: 1) automatically handle the issue and store an associated warning message detailing how the issue was addressed, or 2) stop running entirely or skip the problematic individual and store an error message detailing what the issue was. **Table 4** outlines how different issues are handled in the dashboard, and what actions can be taken to address the issue. **Figure 2** illustrates both how the individual-level residential history, exposure window, and data availability period can overlap and the importance of contiguous, non-overlapping residential history.

Table 4. Description of how different issues are handled by the dashboard, and what actions to take to resolve the issue.

Issue	How Issue is Handled	How to Resolve Issue
Incorrect file format provided	No individuals are processed. Error message is outputted to dashboard.	Ensure that both the residential history and exposure window files are provided in .csv, .xls, or .xlsx format.
Incorrect column names in the residential history and/or exposure window file	No individuals are processed. Error message is outputted to dashboard.	Ensure the below column names are used: - Residential history file: `IndividualID`, `PostalCode`, `StartDate`, `EndDate` - Exposure window file: `IndividualID`, `ExposureStartDate`, `ExposureEndDate`.
Incorrect date format or missing date in the residential history and/or exposure window file	No individuals are processed. Error message is outputted to dashboard.	Ensure there are no missing dates and that all dates in both the residential history and exposure window files are formatted as follows: YYYY-MM-DD.
Differences in the `IndividualID`s included in the residential history and exposure window files	No individuals are processed. Error message is outputted to dashboard.	Ensure that residential history and exposure window files have the same number of `IndividualID`s, and that each `IndividualID` exists in both files.
An `IndividualID` is included more than once in the exposure window file	For each `IndividualID`, the first available exposure window is used, and all other exposure windows are removed. Individual is processed. Warning message is stored in log file.	Ensure that each `IndividualID` appears exactly once in the exposure window file.
Exposure window end date occurs before start date	Individual is not processed. Error message is stored in log file.	Ensure that for each exposure window, the end date occurs after the start date.
Exposure window is less than 1 year in length	Individual is not processed. Error message is stored in log file.	Ensure that each exposure window is at least 1 year in length.

Exposure window starts before or ends after the data availability period	If start date occurs before 2010, set to January 1, 2010. If end date occurs after 2023, set to December 31, 2023. Handling is dependent on length of revised exposure window: - ≥ 1 year: Individual is processed. Warning message is stored in log file. - < 1 year: Individual is not processed. Error message is stored in log file.	Ensure that the start and end dates for each exposure window occur between 2010-2023 and that the exposure window is at least 1 year in length.
Exposure window starts before or ends after the provided residential history	If the exposure window start date occurs before provided residential history, set to earliest available residential start date. If the exposure window end date occurs after the provided residential history, set to last available residential end date. Individual is processed with revised exposure window. Warning message is stored in log file.	Ensure that the residential history for an individual includes residential information for the entire exposure window.
Gap in residential history	Handling is dependent on length of gap in residential history: - < 90 days: The gap is split evenly into two periods, and the first period is assigned to the pre-gap postal code and the second period is assigned to the post-gap postal code. Individual is processed. Warning message is stored in log file. - ≥ 90 days: Individual is not processed. Error message is stored in log file.	Ensure there are no gaps in the residential history for an individual, or if there are gaps, ensure that the gaps are less than 90 days in length.
Overlap in residential history	Handling is dependent on length of overlap in residential history: - < 90 days: The overlapping period is split evenly in two periods, and the first period is assigned to the first postal code, and the second period is assigned to the second postal code. Individual is processed. Warning message is stored in log file. - ≥ 90 days: Individual is not processed. Error message is stored in log file.	Ensure that an individual's residential history is contiguous and non-overlapping, or if there are overlaps, ensure that the overlaps are less than 90 days in length.
Residential history end date occurs before start date	Individual is not processed. Error message is stored in log file.	For all postal codes in the residential history, ensure the start date occurs before the end date.
Invalid or incorrectly formatted postal code in residential history	Individual is not processed. Error message is stored in log file.	Ensure all postal codes included in the residential history file are correctly formatted (6 digits, no spaces) and located in one of the 13 provinces and territories in Canada.

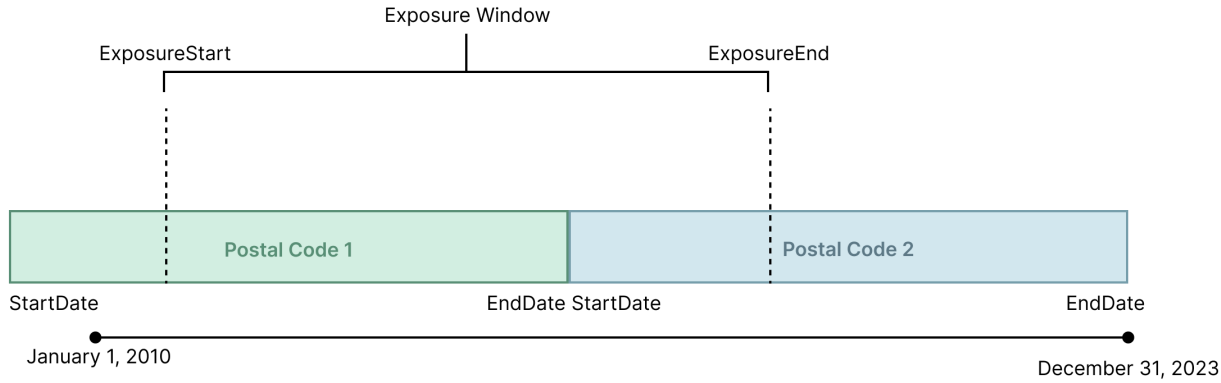


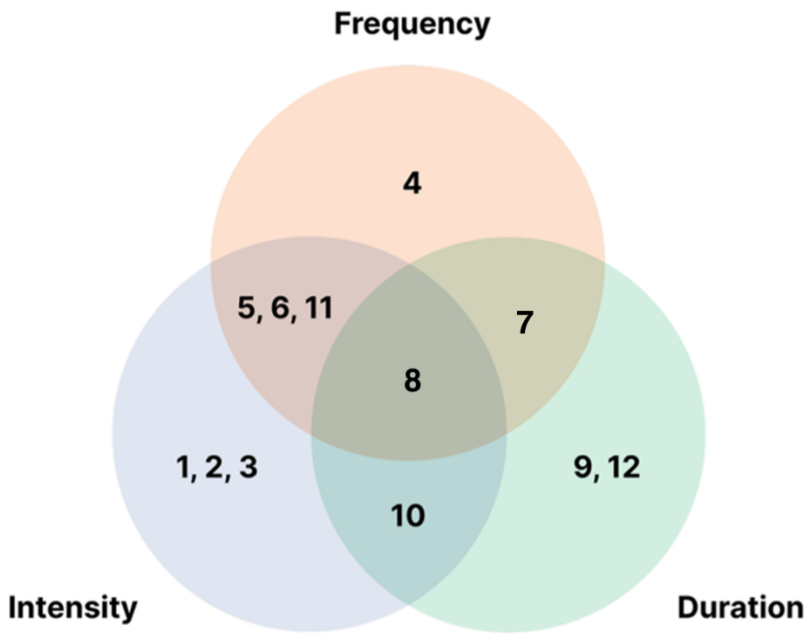
Figure 2. A conceptual diagram illustrating how an individual's residential history can overlap with the exposure window and the data availability period (2010-2023). No overlaps or gaps should exist between postal codes in an individual's residential history. The exposure window must fall within the available residential history and data availability period.

Outputs

There are four outputs provided by the dashboard: **1)** a .csv file with the MultiWiSE metrics for all individuals, **2)** figures illustrating each individual's $PM_{2.5}$ exposure profile, **3)** a .csv file with the weekly estimates of total, WFS, and non-WFS $PM_{2.5}$ and identification of WFS episodes for each individual, and **4)** a log file (.csv) with the error and warning messages for each individual. Each output is described in additional detail below.

1. File of Individual-Level MultiWiSE Metrics

The dashboard will generate a .csv file (MultiWiSE_Metrics.csv) with 12 MultiWiSE metrics for each individual, which characterize the frequency, intensity, and duration of episodic WFS exposure over the provided exposure window (**Figure 3**). In addition, the file will include values for 5 additional variables: the Cumulative Total $PM_{2.5}$ (mg/m^3), Average Total $PM_{2.5}$ (µg/m^3), Cumulative Non-WFS $PM_{2.5}$ (mg/m^3), Average Non-WFS $PM_{2.5}$ (µg/m^3), and Average Counterfactual Weekly $PM_{2.5}$ (µg/m^3). **Table 5** provides an example of the MultiWiSE metrics and variables produced by the dashboard.



Metric	Name	Unit	Definition
1	Cumulative WFS $PM_{2.5}$	mg/m³	Total sum of WFS $PM_{2.5}$ over the multiyear period
2	WFS Fraction	%	Fraction of cumulative total $PM_{2.5}$ exposure from WFS
3	Average WFS $PM_{2.5}$	µg/m³	Average WFS $PM_{2.5}$ across all WFS-impacted weeks
4	Any WFS	# weeks	Total count of WFS-impacted weeks
5	WFS $PM_{2.5}$ > 5 µg/m³	# weeks	Count of WFS-impacted weeks where average WFS $PM_{2.5}$ exceeded 5 µg/m³
6	Total $PM_{2.5}$ > 25 µg/m³	# weeks	Count of WFS-impacted weeks where average total $PM_{2.5}$ exceeded 25 µg/m³
7	WFS Episodes	# episodes	Total count of WFS episodes
8	Severe Episodes	# episodes	Count of severe WFS episodes
9	Longest Episode	# weeks	Length of the longest WFS episode
10	Worst Episode	µg/m³	Average WFS $PM_{2.5}$ during the WFS episode with the highest average WFS $PM_{2.5}$
11	WFS from Severe Episodes	%	Fraction of cumulative WFS $PM_{2.5}$ from severe WFS episodes
12	Average Recovery	# weeks	Average time between episodes, including periods before the first and after the last episode
	Cumulative Total $PM_{2.5}$	mg/m³	Total sum of total (all source) $PM_{2.5}$ over the multiyear period
	Average Total $PM_{2.5}$	µg/m³	Average total $PM_{2.5}$ across all weeks in the exposure window
	Cumulative Non-WFS $PM_{2.5}$	mg/m³	Total sum of non-WFS $PM_{2.5}$ over the multiyear period
	Average Non-WFS $PM_{2.5}$	µg/m³	Average non-WFS $PM_{2.5}$ across all weeks in the exposure window
	Counterfactual Weekly $PM_{2.5}$	µg/m³	The counterfactual value, calculated as the median of total $PM_{2.5}$ during weeks with a modified Z-score between -2 and 2

Figure 3. The 12 MultiWiSE metrics, and 5 additional variables, that capture the frequency, intensity, and duration of an individual's exposure to wildfire smoke (WFS) over the duration of the provided exposure window. The 'Counterfactual Weekly $PM_{2.5}$' represents the average counterfactual value across all postal codes in an individual's residential history.

Table 5. Example of the 'MultiWiSE_Metrics.csv' file with the 12 MultiWiSE metrics and 5 additional variables for two individuals. The units and definitions of the metrics can be found in Figure 3.

Individual ID	1_Cumulative _WFS_PM25	2_WFS_ Fraction	3_Average_ WFS_PM25	4_Any_ WFS	5_WFS_PM25 _exceeds_5	6_Total_PM25 _exceeds_25	7_WFS_ Episodes	8_Severe_ Episodes	9_Longest _Episode	10_Worst _Episode	11_WFS_ from_Severe _Episodes	12_ Average_ Recovery	Cumulative_ Total_PM25	Average _Total_ PM25	Cumulative _NonWFS_ PM25	Average_ NonWFS _PM25	Average_ Counterfactual _Value
Person1	2.71	11.33	12.47	31	22	4	5	2	8	31.42	51.8	72.71	23.89	7.31	21.18	6.88	6.12
Person2	3.24	13.79	13.92	34	25	6	6	3	9	34.85	58.6	75.24	26.13	8.06	22.89	7.18	6.42

2. Figures of Individual-Level Exposure Profiles

The dashboard generates a series of figures for each individual, as shown in **Figure 4**, including:

A) Cumulative $PM_{2.5}$ exposure over the provided exposure window, separated into total, wildfire smoke (WFS), and non-WFS components.

B) Weekly average total, WFS, and non-WFS $PM_{2.5}$ over the provided exposure window, with identification of WFS episodes and severe WFS episodes.

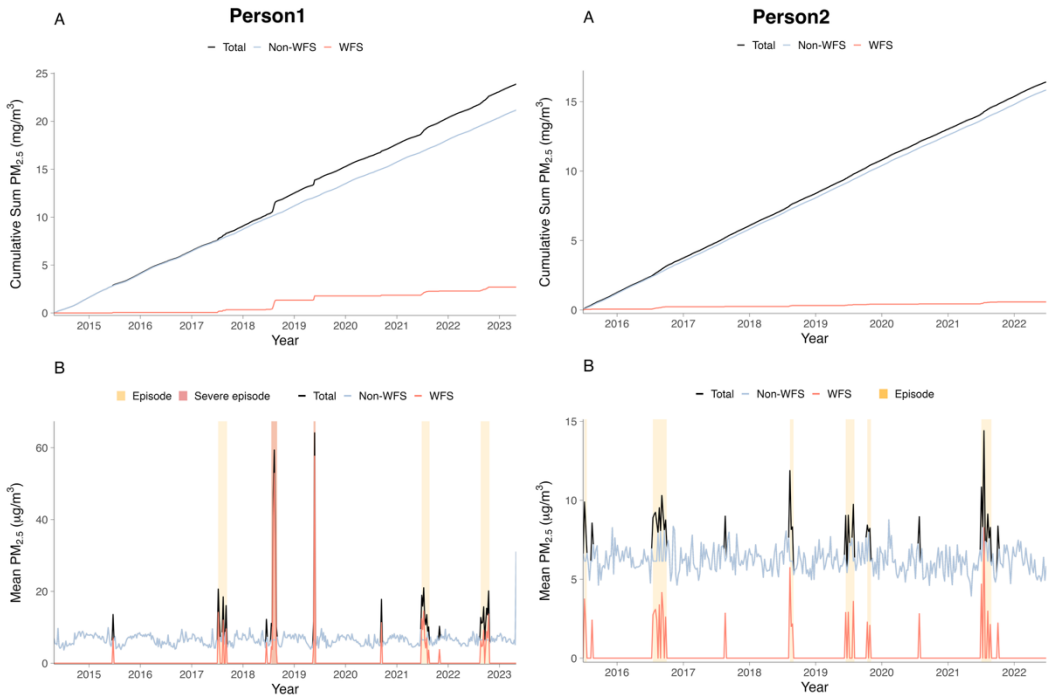


Figure 4. Example of provided figures. The cumulative (top) and average (bottom) weekly total, wildfire smoke (WFS), and non-WFS $PM_{2.5}$ for each individual.

3. File of Individual-Level Weekly $PM_{2.5}$ Estimates

The dashboard will generate a .csv file (Weekly_PM25_Estimates.csv) with individual-level data on the weekly $PM_{2.5}$ estimates during the defined exposure windows. This file includes information on weekly total, WFS, and non-WFS $PM_{2.5}$, as well as identification of WFS episodes. These estimates are used to generate the MultiWiSE metrics and figures. A list of all variables provided can be found in **Table 6**, and an example of the outputted .csv can be found in **Table 7**.

Table 6. Definition of variables provided the 'Weekly_PM25_Estimates.csv' file

Variable	Unit	Description
IndividualID	N/A	A unique identifier for a given individual
PostalCode	N/A	6-digit alphanumeric Canadian postal code, indicating the postal code that the individual lived in during a given week. For weeks where an individual moved postal codes, the postal code with the most overlapping days is used.

epiweek	N/A	The week number the exposure data corresponds to. The first week in the exposure window will be week 1. Weeks are represented as epidemiological weeks, which are standardized to start on a Sunday and end on a Saturday.
n_days	days	The number of days in the epiweek that had data available for analysis. A value of 7 indicates no missing data. Incomplete epiweeks (<7 days) that occur at the beginning and end of the exposure window are dropped and not included in analysis.
epiweek_start_date	weeks	The date on which the epiweek started (always a Sunday), shown in date format YYYY-MM-DD.
epiweek_end_date	weeks	The date on which the epiweek ended (always a Saturday), shown in date format YYYY-MM-DD.
weekly_pm25_avg	$\mu g/m^3$	The weekly average of total $PM_{2.5}$
weekly_wfs_pm25_avg	$\mu g/m^3$	The weekly average of WFS $PM_{2.5}$
weekly_non_wfs_pm25_avg	$\mu g/m^3$	The weekly average of non-WFS $PM_{2.5}$
weekly_pm25_sum	$\mu g/m^3$	The weekly sum of total $PM_{2.5}$
weekly_wfs_pm25_sum	$\mu g/m^3$	The weekly sum of WFS $PM_{2.5}$
weekly_non_wfs_pm25_sum	$\mu g/m^3$	The weekly sum of non-WFS $PM_{2.5}$
episode_id	N/A	Unique identifier for each WFS episode. All weeks in a given WFS episode will have the same identifier. A WFS episode is defined as either (1) two or more WFS-impacted weeks separated by no more than three non-WFS-impacted weeks, or (2) a single week where the sum of WFS $PM_{2.5} > 250$ mg/m^3.
severe_episode_id	N/A	Unique identifier for each severe WFS episode. All weeks in a given severe episode will have the same identifier. A severe episode is defined as any WFS episode that includes a week with a sum of WFS $PM_{2.5} > 250$ mg/m^3.

Table 7. Example of the 'Weekly_PM25_Estimates.csv' file with weekly $PM_{2.5}$ estimates and identification of WFS episodes for each individual.

IndividualID	PostalCode	epiweek_index	num_days	epiweek_start_date	epiweek_end_date	weekly_pm25_avg	weekly_wfs_pm25_avg	weekly_non_wfs_pm25_avg	weekly_pm25_sum	weekly_wfs_pm25_sum	weekly_non_wfs_pm25_sum	episode_id	severe_episode_id
Person1	Y1A0J4	1	7	2014-04-27	2014-05-03	6.89	0.00	6.89	20.68	0.00	20.68	0	0
Person1	Y1A0J4	2	7	2014-05-04	2014-05-10	5.41	0.00	5.41	37.90	0.00	37.90	0	0
Person1	Y1A0J4	3	7	2014-05-11	2014-05-17	5.51	0.00	5.51	38.56	0.00	38.56	0	0
Person1	T8V8K	4	7	2014-05-18	2014-05-24	17.79	11.31	6.48	124.56	79.20	45.36	1	0
Person1	T8V8K	5	7	2014-05-25	2014-05-31	20.13	13.65	6.48	140.93	95.57	67.75	1	0
…													
Person2	Y1A0J4	1	7	2014-06-08	2014-06-14	4.17	0.00	4.17	29.16	0.00	29.16	0	0
Person2	Y1A0J4	2	7	2014-06-15	2014-06-21	4.36	0.00	4.36	30.51	0.00	30.51	0	0
Person2	C0A0A1	3	7	2014-06-22	2014-06-28	5.15	0.00	5.15	36.07	0.00	36.07	0	0
Person2	C0A0A1	4	7	2014-06-29	2014-07-05	9.00	2.85	5.66	62.99	19.97	43.02	0	0
…													

4. Log file with warning and error messages

The dashboard will generate a .csv file (Warning_Error_Log.csv) that logs all the error and warning messages that occurred when processing each individual. For an individual who was processed without issue, the log file will contain a message stating "No issues." This log file can be used to identify and resolve issues with the user-provided files. An example of the outputted .csv can be found in **Table 8**.

Table 8. Example of the 'Warning_Error_Log.csv' file that logs the warning and error messages for each individual.

IndividualID	Message
Person1	No issues
Person2	Error: Unable to process individual. In the residential history, the provided end date for T8V8K occurs prior to the start date. For all postal codes, please ensure the start date occurs before the end date, and that the residential history is contiguous and non-overlapping.
Person3	No issues
Person4	Warning: The provided start date of the exposure window is outside the data availability window (2010 – 2023). The exposure window start date was automatically set to January 1, 2010.

Supplemental

Data Source

The estimates of wildfire smoke (WFS) and non-WFS $PM_{2.5}$ come from the application of the MultiWiSE approach (Steps 1-5 in **Figure 1**) to total (i.e., all-source) $PM_{2.5}$ estimates from the Canadian Optimized Statistical Exposure Model (CanOSSEM). CanOSSEM is a machine learning model that estimates daily average total $PM_{2.5}$ concentrations across all populated regions of Canada using multiple predictor variables. More information about CanOSSEM can be found at: https://www.sfu.ca/fhs/canossem.

The Data Repository

The postal code-level estimates of weekly total, WFS, and non-WFS $PM_{2.5}$, which are used to generate the individual-level exposure profiles and MultiWiSE Metrics, are stored in Dropbox and are open access. However, to use the dashboard, users do **not** need to directly access the Dropbox. Each file in the repository contains 14-year data (2010-2023) on the weekly averages and sums of total, WFS, and non-WFS $PM_{2.5}$ concentrations for a given 6-character postal code.

Link to data repository:
https://www.dropbox.com/scl/fo/ynbextn32g93lj3xirh6s/AJauEcPHlBlOX6Cl3lcEcec?rlkey=1se5jryxdzlsqtlsqp1nclil9&st=99ya8gwo&dl=0

Support

If you require further support or have questions regarding this dashboard, please contact:

Stephanie E. Cleland, PhD | email: stephanie_cleland@sfu.ca
Sarah B. Henderson, PhD | email: sarah.henderson@bccdc.ca

Acknowledgement

This README and dashboard were the joint effort of Olivia Qiong (Research Assistant, BCCDC and VCHRI), Stephanie Cleland (Assistant Professor, SFU and VCHRI), and Sarah Henderson (Scientific Director, BCCDC and UBC). The research team acknowledges the contributions of Naman Paul and Michael Brauer to the project.